Course 4: Python for Data Science & Machine Learning

Data Engineering, Feature Pipelines & ML Deployment Handbook

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1. Vectorized Computing with NumPy & Pandas

Python loops introduce severe GIL and interpreter overhead when processing large numerical matrices. NumPy vectorization executes operations at compiled C/BLAS speeds.

Avoid chained indexing (df['col'][idx]) in Pandas; use df.loc and df.iloc to guarantee view-versus-copy deterministic updates.

```
import numpy as np
import pandas as pd

# Vectorized feature transformation
df['log_income'] = np.log1p(df['income'].clip(lower=0))
df['scaled_age'] = (df['age'] - df['age'].mean()) / df['age'].std()
```

2. Cross-Validation & Preventing Data Leakage

Data leakage occurs when information from outside the training dataset is inadvertently used to create the model. Preprocessing transforms (imputation, scaling) must be fitted strictly on the train folds.

Use StratifiedKFold for classification tasks where class balance is imbalanced to ensure equal target distribution across folds.

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('classifier', RandomForestClassifier(n_estimators=100, random_state=42))
])
```

3. Model Evaluation Metrics

Accuracy is misleading for imbalanced datasets (e.g. fraud detection with 99.9% negative cases). Instead, monitor Precision, Recall, F1-Score, and ROC-AUC curve distributions.

Model explainability via SHAP (SHapley Additive exPlanations) values helps diagnose feature importance for regulatory compliance.

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